Talha Hussain Khan

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EDUCATION

University of North Carolina Wilmington **Wilmington, USA**
M.S. in Computer & Information Sciences *Jan 2026 – Current*
Sir Syed University of Engineering & Technology **Karachi, Pakistan**
B.S. in Computer Engineering *Aug 2020 – Nov 2024*

- **GPA:** 3.57/4.00

HONORS AND ACHIEVEMENTS

- **2nd Position:** Earned 2nd place in the Hello GPT-4o AI Challenge by LabLab.ai, competing against 168 teams. [Link]
- **Awarded 2nd position** for my poster presentation at the 40th All Pakistan IEEEP Students' Seminar 2025. [Link]
- **FYP Nominated:** Selected by the university to present at ITCN Asia Expo and advanced to the 2nd stage in FICS. [Link]
- **Merit Scholarship Recipient:** Awarded for outstanding academic performance during bachelor's degree, ranking in the **top 5%** of the batch.
- **AIESEC Selection:** First student from the university to be selected for AIESEC, a global youth organization that provides leadership development and cross-cultural internships, enhancing global impact and personal growth**.**

RESEARCH EXPERIENCE

3D Mapping of Buildings for Inspection | *Research Project in collaboration with RoadGauge Ltd* | **Final Year Project** [Link]
[Preprint] *Nov 2023 – Sept 2024*

- Spearheaded development of cost-effective alternative to LIDAR-based building inspection: designed end-to-end 3D mapping system integrating IMU sensors, Android SDK, and Structure from Motion (SfM) algorithms, reducing equipment costs by ~80% while maintaining millimeter-level accuracy
- Led backend architecture, data collection framework, Arduino-based hardware automation for consistent video capture, and computer vision pipeline for 3D model generation from image sequences
- Technologies: Arduino, Android SDK, OpenCV, SfM, Computer Vision, IMU Sensors, Python, Firebase

EXPERIENCE

Meezan Bank **Karachi, Pakistan**
IT Officer *Oct 2025 – Dec 2025*

- Monitored and maintained ADC and digital banking systems to ensure smooth and reliable daily operations.
- Reported and resolved technical issues promptly, reducing service interruptions and improving customer experience.
- Created automation scripts and work flows to simplify routine tasks and improve operational efficiency.
- Supported the deployment of system updates and digital banking enhancements.

Linuro Systems **Karachi, Pakistan**
Junior Dot Net Developer *Jan 2025 – Sep 2025*

- Developed and maintained software modules using .NET technologies, improving system reliability and performance.
- Supported testing and debugging tasks, helping reduce application errors and enhance user experience.
- Collaborated with senior developers to implement new features, contributing to timely project delivery.

RoadGauge Ltd **(Remote) - England, UK**
Software Development Engineer (Backend) *Jan 2023 – Mar 2024*

- Upgrade and maintain the app by adding new features, functionalities into one single code base using Java.
- Focused on backend development, specifically in building a custom report generation system. Utilized Handlebars for report templating and successfully built a custom report generation system that reduced the time between report request and report delivery by 90%.
- Contributed to our team's efforts in integrating the Stripe payment gateway and deploying APIs on the Google Cloud Platform using Node.js and Python.
- Deployed APIs on the Google Cloud Platform, contributing to improved system functionality.
- Performed data labeling and annotation for computer vision datasets using CVAT, enhancing model accuracy.

INTERNATIONAL HACKATHONS & COMPETITIONS

AI in Education: Lang Anki Cards on Steroids [GitHub]**|**Hello GPT-4o AI Challenge

- Developed an advanced version of Anki flashcards for educational purposes using GPT-4o.
- Focused on enhancing language learning through AI-driven flashcards.

- Utilized AI for generating and optimizing educational content.
- Awarded **2nd place** in the challenge for innovative use of AI in education.

In-Car AI Agents [GitHub] **|** Edge Runners 3.2 Hackathon

- Developed an offline AI system for cars using LLaMA 3.2, enabling voice commands for functions like air conditioning and music playback without internet.
- Deployed edge-based AI with pre-trained voice recognition for real-time responses, enhancing driver experience in low-connectivity areas.
- Planned enhancements include offline navigation and lane detection for increased safety and autonomy.

Data Science for Bioengineering Program by UC Irvine: Gained hands-on experience in applying data-driven methods to solve biological and biomedical challenges using real-world datasets.

Coding/Problem Solving Events: Harvard CS50x Puzzle Day'25 (9/9 puzzles), MIT Informatics Tournament Winter'25 (75/500+ teams), CALICO Fall'24- UC Berkeley (248/865 teams), Meta Hacker Cup'24

PROJECTS

Ride-Sharing App | *AWS Cloud Project* [Link]

- Deployed full-stack ride-sharing application using AWS services including Amplify, Cognito, Lambda, and DynamoDB.
- Implemented serverless backend with API Gateway and Lambda functions for real-time ride matching.
- Developed secure user authentication system using AWS Cognito and IAM roles.
- Technologies: AWS Suite (CodeCommit, Amplify, Lambda), API Gateway, DynamoDB

Soccer Player Manager | *Software Engineering Course Project* [Link]

- Developed a web application for managing soccer players, allowing users to add, update, and view player profiles.
- Implemented the backend using PHP and MySQL, with a frontend built using HTML, CSS, and JavaScript.
- Designed and integrated a user-friendly interface for smooth navigation and player management.
- Technologies: PHP, JavaScript, HTML, CSS, MySQL

Snake Game - MIPS Assembly | *Low-Level Programming Project*

- Engineered Snake game in MIPS assembly language, implementing complete graphics rendering, collision detection, and real-time keyboard input system.
- Technologies: MIPS Assembly, Bitmap Display, Keyboard Input/Output

IoT Street Infrastructure | *Hardware Projects*

- Engineered a smart street lighting prototype integrating motion and ambient light sensors, reducing energy consumption through automated brightness control.
- Developed custom metal detection system using electromagnetic induction and analog circuit design.
- Technologies: Microcontrollers, Sensors, Electromagnetic Induction, Circuit Design

TEACHING AND VOLUNTEER EXPERIENCE

Volunteer Teaching *Jan 2021 – Current*

- Conducted computer programming and data structures and algorithms sessions for 50+ underprivileged students, strengthening their programming skills, collaborating with volunteer organizations like Iqra Fund (USA). [Link]
- Mentored and judged at PEC Generative AI Hackathon & WordSprint Developers Hackathon. [Link]
- Provided private tutoring & study group moderation for secondary, Cambridge O-Level, and A-Level students in Math, Physics, and CS.
- Delivered a Meta Hacker Cup webinar at ACM SSUET. [Link]

RHA | *Volunteer* *Oct 2022 – Current*

- Led weekly food/ration & winter drives for underserved communities; received "Consistent Volunteer Award."

The Citizens Foundation | *Volunteer* *Feb 2024 – Current*

- Advocated for education & fundraising for underprivileged children in Pakistan via TCF's Baghbaan program.

SKILLS

Programming: Python, JavaScript, SQL, Node.js, MATLAB, C, Java, Verilog (basic)
Tools: Git, Postman, Android Studio, AWS, GCP, Firebase, Open MP & MPI, CVAT, CloudCompare, 3dsMax, PuTTY, AutoCAD(basic), MARS MIPS simulator, PSpice, Multisim, Electronic Workbench